

# ML Challenge 2026: Business Entity Resolution Solution Template

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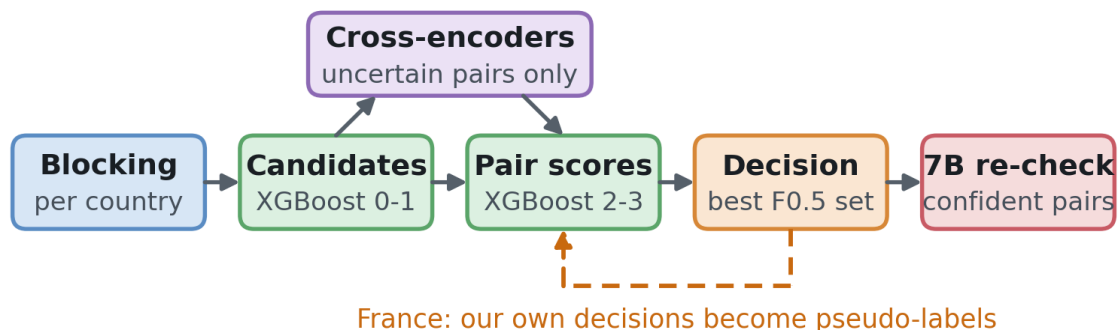
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## 1. Executive Summary

Our submission, which we call Composite B, scores 0.990879 macro F0.5 on the public leaderboard. On our own local validation set, a held-out quarter of the labelled US and India data, the same pipeline scores 0.9913. That set has no French records, because France has no labels.

The pipeline makes three passes. Blocking looks up candidate records for each Source-1 (S1) business one country at a time, and a gradient-boosted classifier cuts them down to 3.70 per S1 while keeping 99.1% of the true pairs. More gradient-boosted models then score every candidate pair. On the pairs they are least sure about, fine-tuned multilingual cross-encoders help: e5, bge, and Qwen2.5-1.5B and 7B with LoRA. Finally, for each S1 we pick the set of records with the highest expected F0.5. There is no global threshold.

France was the hard part. It appears only in the test set, so we had nothing to train or validate on there. We self-trained on our own French decisions, with guards and cross-fitting so the model would not simply confirm its own mistakes. We also had a Qwen2.5-7B model re-read the predictions the pipeline was already certain about, and that second reading removed a group of French decoys that string features could not separate.



**Figure 1.** The pipeline, from records (left) to matches (right). The dashed line is the French self-training loop (Section 2.2).

## 2. Methodology

### 2.1 Problem Analysis

Each S1 business has between 0 and 11 copies in the Source-2 and Source-3 files, 3.46 on average, and about 5.6% have none at all. The metric is F0.5 averaged over S1, so a wrong merge costs twice as much as a missed copy. An S1 with no copies also scores zero as soon as we predict anything for it. That pushed the whole design towards precision.

The copies are noisy. In the labelled data we found case and accent changes, legal forms dropped or rewritten (in France, SARL, SAS or EURL changes in about 23% of copies), reordered or appended words, acronyms, typos, shortened street types ("Rue" to "R."), reordered addresses, changed house numbers and empty addresses. Most of our blocking keys and features target one of these edits directly.

The test set also contains decoys. It has about 23% more records per S1 than the training set but the same number of true matches, so the extra records are look-alikes: a different business word at the same address, a nudged house number, or the same name somewhere else. Between 46% and 54% of S1 share their name with another S1, so the address usually settles the match. We therefore let candidates compete within each S1, and each record can belong to at most one S1.

France is harder still. Its names are generic, a city plus a category such as "lille club sarl", and each French decoy we removed shares its name with a median of 43 other S1. About 11% of French S1 share their exact address with another S1, and 32% of French addresses contain a department or region name.

We noticed early how much this mattered. Our first uploads scored 0.976 to 0.980 on the leaderboard, while our local validation said 0.984 to 0.989. The only thing the leaderboard had that our validation lacked was France. Working backwards put France at around 0.93, and most of what we did afterwards was aimed at it.

### 2.2 Solution Strategy

**Approach Type:** Hybrid. Blocking generates candidates, three stages of gradient-boosted classifiers score the pairs, cross-encoders read the uncertain pairs, a per-S1 decision picks the final set, rules clean up known patterns, and an LLM re-checks the confident predictions (Figure 1).

**Core Innovation:** three ideas carried most of the gain.

*Self-training for a country without labels.* We pseudo-label French pairs from our best model's final decisions. A pair it kept with a calibrated probability of at least 0.9 becomes a positive, one it rejected at 0.05 or below becomes a negative, and everything in between stays unlabelled. Two guards stop the model from reinforcing its own errors: where our rules decided a population, the rules win, and empty-address records keep their first-round labels. The labels are also cross-fitted by S1 group, so a French pair is never scored by a model trained on its own label. The first round added 0.00046 on the leaderboard, and the second round, together with the French decision layers, another 0.00015.

*A decision that optimises the metric directly.* For each S1, a small dynamic programme over the calibrated candidate probabilities finds the prediction set with the highest expected F0.5. On our local holdout this beat the best global threshold by 0.000048 (95% interval 0.000007 to 0.000091).

*A second opinion on predictions we were sure about.* 94.5% of our final predictions have a stage-1 probability above 0.99, so no cross-encoder ever looked at them. A LoRA-tuned Qwen2.5-7B reads them again, and we drop the ones it rejects with a logit below  $-6$ , a cut-off we fixed on labelled data first. It dropped 840 French predictions. On labelled data, only 8% of predictions rejected that strongly are real matches.

*How we decided what to keep.* For US and India we held out a fixed 25% of the training S1 (549,699 S1) and never trained on it. A component stayed only if it gained on that holdout in a paired bootstrap, and ties went to the simpler option. Rules added late also had to gain on both halves of the holdout. France has no labels, so there we relied on label-free checks, on estimates corrected with the 7B model's judgements, and on leaderboard submissions that changed France alone.

### 3. Candidate Generation (Blocking)

**Blocking keys used:** All searches run separately per country, so IDF weights and nearest neighbours never mix countries. Records without a country are searched in every partition. Table 1 lists the three views. The token view does most of the work. The names-only view covers records whose address is empty or very short, and the repairs fix names before any matching.

**Table 1.** Blocking views.

| view            | how it matches                                                                                                                                                           | what it catches                                                                                 |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Token view      | IDF-weighted overlap on name + address, both ways: each S1's top 40 records and each record's top 8 S1. A pair is kept if it is in the S1's top 15 or the record's top 4 | most copies. A (house number, street word) key that skips street types handles French addresses |
| Names-only view | character 4-grams; single-token names of 8 or more letters                                                                                                               | empty or short addresses, garbled names, domains and handles                                    |
| Repairs         | domain names split into the country's words; OCR digits in names fixed; an Indic-to-Latin dictionary of 693 entries learned from training pairs                          | web-style names, OCR errors, transliterated names                                               |

**Candidate pairs generated:** Retrieval produces 58.4M pairs on the test set (66.4M on train). A first gradient-boosted scorer (stages 0 and 1) keeps every pair with  $p1 \geq 0.02$ , plus the two best S1 for each record, and we add pairs found by matching acronyms. The candidate file has 6,410,308 pairs, 3.70 per test S1.

**How you ensured true matches were not lost:** We measured recall on the local holdout at every change instead of assuming it. The candidate set contains 99.1% of the true holdout pairs. Most of the 16.5k it misses (out of about 1.9M) are empty-address copies whose name was also changed. Two tighter cuts, keeping only the best S1 per record or raising the threshold to 0.05, lowered holdout F0.5 by 0.000106 and 0.000041, so we kept the looser cut.

### 4. Matching Model

**Features used:** All statistics are fitted per country and there is no country feature, so nothing in the model is tied to US or India. Table 2 groups the features and says why each group is there.

**Table 2.** Feature groups.

| group                    | main features                                                                                                                                                                                     | why                                                                          |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Name                     | fuzzy ratios (token-sort, token-set, partial, Jaro-Winkler); token Jaccard and containment; IDF-weighted cosine; legal-form agreement; counts of unexplained tokens; learned look-alike word odds | a copy keeps the rare words of a name, while a decoy swaps one business word |
| Address and house number | house-number equality, nudges of 1 to 10, digit edits and suffixes; street overlap; city and state; whole-address similarity; empty and PO-box flags                                              | when two businesses share a name, the address tells them apart               |
| Context                  | rank in each blocking view; the gap to the S1's best candidate and to rival S1; name-collision counts; per-S1 aggregates; cluster support; the cross-encoder score                                | a pair is judged against its competitors, not on its own                     |

**Model type:** The pair classifiers are four XGBoost stages. Each is trained out-of-fold over three S1 groups, so every stage learns from honest scores of the one before it.

- **Stage 0** removes about 80% of pairs cheaply.
- **Stage 1** produces the probability  $p_1$ , which sets the candidate cut and decides which pairs go to the cross-encoders.
- **Stage 2** adds the cross-encoder score, per-S1 and cluster features, and the French pseudo-labelled rows. Its output is calibrated with isotonic regression.
- **Stage 3** re-scores each pair against the other candidates of its S1.

The cross-encoders only see the uncertain band: pairs with  $0.02 \leq p_1 \leq 0.99$ , which is 1.49M on the test set. Each is trained as three out-of-fold models on the text "name ; address" of both records, cut to 96 tokens. We average their z-scored logits into one stage-2 feature, because given separate scores, stage 2 extrapolated on pairs where the models disagree, and they disagree about four times as often in France.

Table 3 lists the models. French self-trained versions of e5-large and bge are also in the mix, and e5-base (278M, MIT) was used only for the first French teacher. The 7B model is no more accurate than a self-trained e5-large; both score 0.944. It earns its place in two ways: counted twice in the US/India mix it adds diversity, and it gives an independent reading of confident pairs in the re-check.

**Table 3.** Cross-encoders. Band AUC is measured on labelled pairs from the local holdout; stage-1  $p_1$  alone reaches 0.930 on the same pairs.

| model                            | size        | licence    | training       | band AUC      |
|----------------------------------|-------------|------------|----------------|---------------|
| multilingual-e5-small / e5-large | 118M / 560M | MIT        | full fine-tune | 0.939 (large) |
| bge-reranker-v2-m3               | 568M        | Apache-2.0 | full fine-tune | 0.942         |
| Qwen2.5-1.5B                     | 1.5B        | Apache-2.0 | LoRA           | 0.938         |

| model      | size | licence    | training                                             | band<br>AUC |
|------------|------|------------|------------------------------------------------------|-------------|
| Qwen2.5-7B | 7.6B | Apache-2.0 | LoRA (r 16, alpha 32), three H100s for about 2 hours | 0.944       |

**Threshold selection method:** There is no single threshold. The calibrated stage-3 probabilities go into the expected-F0.5 selection from Section 2.2. Three settings were tuned on the local holdout: a logit shift of +0.2, a further −0.3 for records that four or more S1 compete for, and 0.01 expected true matches outside the candidate set. Each record is then assigned to its highest-scoring S1.

Rule layers follow. Acronym joins and per-source caps apply everywhere, and France also gets an exact-address copy rule (99.99% precise on US and India), a cross-commune drop, a look-alike word-swap drop, and the same set selection run on France's own probabilities. The last step is the 7B re-check at −6. That cut-off had the largest gain on the local holdout (+0.000037), positive in both halves and in 99.7% of random subsets.

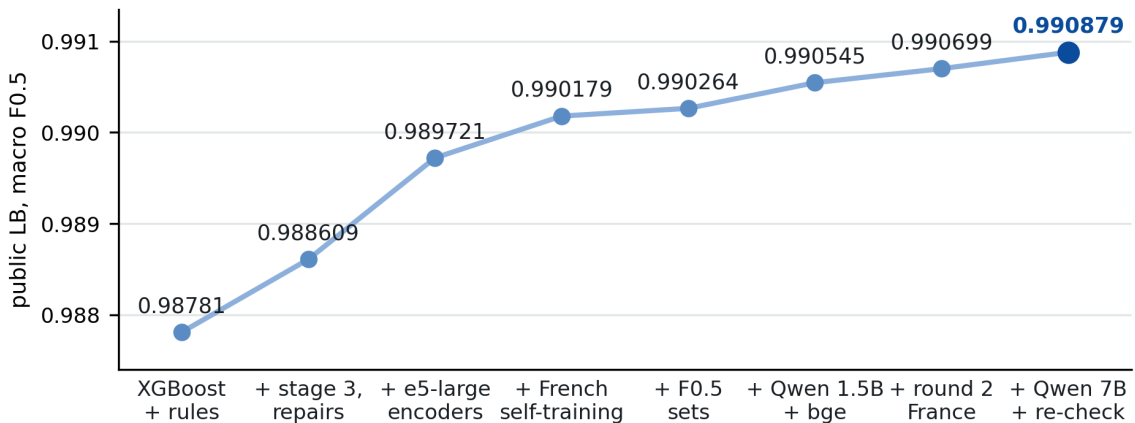
## 5. Results & Error Analysis

**F\_0.5 Score (macro):** Table 4 compares the official leaderboard score with our local validation. The local number is higher because France cannot be part of it, and most of the gap comes from France. The private leaderboard, which decides the final ranking, had not been published when we wrote this.

**Table 4.** Macro F0.5.

| evaluation                                                         | macro F0.5                                                      |
|--------------------------------------------------------------------|-----------------------------------------------------------------|
| public leaderboard, official (US, India and France)                | <b>0.990879</b>                                                 |
| local validation: labelled holdout, US and India only (549,699 S1) | 0.9913 (US 0.9911, India 0.9916); precision 99.9%, recall 97.5% |

Figure 2 shows how the leaderboard score grew over our submissions. The largest steps were the e5-large cross-encoders (+0.0011) and the blocking repairs together with stage 3 (+0.0008). French self-training added 0.00046, and the last step, the 7B model in the mix plus the re-check, added 0.00018.



**Figure 2.** Public leaderboard score after each step. The last point is the submission in this package.

**Common false positives (wrong merges):** The most expensive errors were generic-name decoys in France: the same generic name and house number on a different street, such as "lille ecole sarl, 42 rue gutenber" against "lille ecole sarl, 42 q. du wault". On labelled data this pattern is a real match 0.5% of the time when our model rejects it, and 99.7% of the time when it accepts it. String features cannot tell the two apart, but the 7B model can, and removing its 840 French rejects was worth an estimated 0.00011 of the final 0.00018. Smaller groups of errors are business words swapped at the same address ("antenne danse" and "antenne gaz"), very short names that differ by one letter ("osd" and "otd"), and branches of a chain that share a name.

**Common false negatives (missed matches):** Precision on the local holdout is 99.9%, so most of the remaining loss is recall. 69% of it comes from S1 where we found some copies but not all, and 20% from S1 we missed completely. The missed copies are mostly records with an empty address and a changed name, which either never become candidates (16.5k) or get a score near zero (15.2k). Heavy garbles, such as the French "EHPAD" typed as "ehpvd", account for more. None of the recall rules we tried was more than 71% precise, and under F0.5 an added pair must be right about 75% of the time to help, so we did not add any.

## 6. Conclusion

On US and India, a precision-first pipeline with calibrated per-S1 decisions reached 0.9913 on our local validation. On the public leaderboard, which includes France, the submission scored 0.990879. Self-training on our own guarded decisions got France most of the way without any labels. The 7B re-check showed that a second, independent model can catch decoys that feature-based models accept with confidence.

Two lessons stay with us. Self-training helped for two rounds, and a third did not add to them. And on a country without labels, any estimate built from the model's own probabilities needs an independent check, because it shares the model's blind spots.

## Appendix

### A. Code Artefacts

`code/business_entity_resolution/` regenerates both output files from the raw train and test TSVs. Its `README.md` gives the exact commands, the run time of each step, and what each step should print.

`reproduce.sh` is the entry point; with `VARIANT=compositeB` (the default) it runs `src/box/compositeB.sh`. The team package is in `src/ber`, the model chain (XGBoost stages, cross-encoders, self-training labels, decision, rules, the France block) in `src/model_v1`, and the Qwen2.5-7B training, the 7B re-check and the per-country composition in `src/box`. The XGBoost stages need 24 or more CPU cores and 32 GB of RAM, the cross-encoders ran on one 80 GB H100, and Qwen2.5-7B needs three 80 GB GPUs for about two hours. We used Python 3.12 to 3.13, torch 2.11, transformers 5.17, peft 0.21 and xgboost 3.2.0.

Seeds are fixed, and every artifact records the command and git commit that made it. GPU training is not bit-identical across machines (an earlier model rebuilt on other hardware moved from 0.991246 to 0.991261 on the holdout), so a rerun should land within about 0.0001. The `output/` folder holds the

exact bytes we submitted.

## B. Additional Results

**Table B1.** Ablations on the local validation set (labelled US/India holdout), each tested with a paired bootstrap.

| change                                                   | effect on macro F0.5                    |
|----------------------------------------------------------|-----------------------------------------|
| expected-F0.5 set selection instead of a tuned threshold | +0.000048 (0.000007 to 0.000091)        |
| Qwen2.5-7B counted twice in the cross-encoder mix        | India +0.000066 (P 0.998), US +0.000026 |
| 7B re-check, dropping $\text{logit} < -6$                | +0.000037, positive in both halves      |
| tighter candidate cut (best S1 only / $p1 \geq 0.05$ )   | -0.000106 / -0.000041                   |

On a labelled 34% sample of the local holdout, 8.3% of the confident predictions the 7B scored below  $-6$  were real matches, against 92.6% between  $-6$  and  $-2$  and over 99.9% above that. The 7B rejects 0.10% of French predictions, but only 0.004% to 0.007% of US and India ones. The French rate is the same in each third of the French S1, including the third the model never saw labels for, so the re-check is not repeating its training labels.

Two leaderboard controls support these choices. Composite B with the round-3 French model, but without the French decision layers, scored 0.990833, which is 0.000046 lower. Pushing the French 7B drops below  $-6$  scored 0.990875, a tie, so the  $-6$  cut-off was already in the right place.

We also measured and dropped several ideas:

- recall rules, which were 17% to 71% precise;
- a learned blend of all six cross-encoder scores as the decision score (band AUC 0.9575, but 0.001 lower F0.5);
- mining extra drops with a decision tree (+0.000033 on one half of the holdout, -0.000015 on the other);
- breaking ties between empty-address records by copy count (-0.00057);
- renormalising French probabilities per record (-0.000007 to -0.000137);
- other encoders: Qwen3-4B scored below the 7B, and mDeBERTa-v3 and gte-multilingual could not be trained or run reliably.